A RIG-ETALE RIG-SURJECTIVE DIAGONAL FORCES THE FLAT LOCUS OVER THE DVR TO BE EVERYTHING OR NOTHING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a base change of the diagonal properties, an affine completed-tensor-product computation, a closed immersion dichotomy, and a sheaf injectivity argument promoted through pairwise unions of cover members, all inside the delivered span

Editorial note. The label lemma-flat-and-diagonal-rig-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The flat locus closed immersion X' of the flat locus lemma of the same chapter, and base change behaviour of rig-etale and rig-surjective morphisms
- The fibre product presentation of affine formal algebraic spaces over a separated base, and flatness of completed tensor products of flat algebras
- The rig-smooth rig-surjectivity dichotomy for closed immersions from the same chapter

Source reference. Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at proof line(s) 5887; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AQ5: sources/stacks-chow/tags.txt, source lines 13498–13498. Prerequisite pointer invoked at proof line(s) 5846; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN2: sources/stacks-chow/tags.txt, source lines 13425–13425. Prerequisite pointer invoked at proof line(s) 5843; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN3: sources/stacks-chow/tags.txt, source lines 13426–13426. Prerequisite pointer invoked at proof line(s) 5878; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AQ7: sources/stacks-chow/tags.txt, source lines 13500–13500. Prerequisite pointer invoked at proof line(s) 5847; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQS: sources/stacks-restricted/pinned-original-source.tex, source lines 5324–5324. Prerequisite pointer invoked at proof line(s) 5858; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GD6: sources/stacks-restricted/pinned-original-source.tex, source lines 5602–5602. Prerequisite pointer invoked at proof line(s) 5882; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GD8: sources/stacks-restricted/pinned-original-source.tex, source lines 5752–5752. Prerequisite pointer invoked at proof line(s) 5835; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a locally Noetherian formal algebraic space which is locally of finite type over a complete discrete valuation ring A. Let $X' \subset X$ be as in Lemma 0GD8. If $X \to X \times_{Spf(A)} X$ is rig-étale and rig-surjective, then $X' = Spf(A)$ or $X' = \emptyset$.*

Proof. (Aside: the diagonal is always locally of finite type by Formal Spaces, Lemma 0AN2 and $X \times_{\mathrm{Spf}(A)} X$ is locally Noetherian by Formal Spaces, Lemmas 0AQ5 and 0AQ7. Thus imposing the conditions on the diagonal morphism makes sense.) The diagram

$$\begin{array}{ccc} X' & \longrightarrow & X' \times_{\mathrm{Spf}(A)} X' \\ \downarrow & & \downarrow \\ X & \longrightarrow & X \times_{\mathrm{Spf}(A)} X \end{array}$$

is cartesian. Hence $X' \to X' \times_{\mathrm{Spf}(A)} X'$ is rig-étale and rig-surjective by Lemma 0AQS. Choose an affine formal algebraic space U and a morphism $U \to X'$ which is representable by algebraic spaces and étale. Then $U = \mathrm{Spf}(B)$ where B is an adic Noetherian topological ring which is a flat A-algebra, whose topology is the π-adic topology where $\pi \in A$ is a uniformizer, and such that $A/\pi^n A \to B/\pi^n B$ is of finite type for each n. For later use, we remark that this in particular implies: if $B \neq 0$, then the map $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$ is a surjection of sheaves (please recall that we are using the fppf topology as always). Repeating the argument above, we see that

$$W = U \times_{X'} U = X' \times_{X' \times_{\mathrm{Spf}(A)} X'} (U \times_{\mathrm{Spf}(A)} U) \longrightarrow U \times_{\mathrm{Spf}(A)} U$$

is a closed immersion and rig-étale and rig-surjective. We have $U \times_{\mathrm{Spf}(A)} U = \mathrm{Spf}(B \widehat{\otimes}_A B)$ by Formal Spaces, Lemma 0AN3. Then $B \widehat{\otimes}_A B$ is a flat A-algebra as the π-adic completion of the flat A-algebra $B \otimes_A B$. Hence $W = U \times_{\mathrm{Spf}(A)} U$ by Lemma 0GD6. In other words, we have $U \times_{X'} U = U \times_{\mathrm{Spf}(A)} U$ which in turn means that the image of $U \to X'$ (as a map of sheaves) maps injectively to $\mathrm{Spf}(A)$. Choose a covering $\{U_i \to X'\}$ as in Formal Spaces, Definition 0AIM. In particular $\coprod U_i \to X'$ is a surjection of sheaves. By applying the above to $U_i \coprod U_j \to X'$ (using the fact that $U_i \amalg U_j$ is an affine formal algebraic space as well) we see that $X' \to \mathrm{Spf}(A)$ is an injective map of fppf sheaves. Since X' is flat over A, either X' is empty (if U_i is empty for all i) or the map is an isomorphism (if U_i is nonempty

for some i when we have seen that $U_i \to \mathrm{Spf}(A)$ is a surjective map of sheaves) and the proof is complete. $\square$